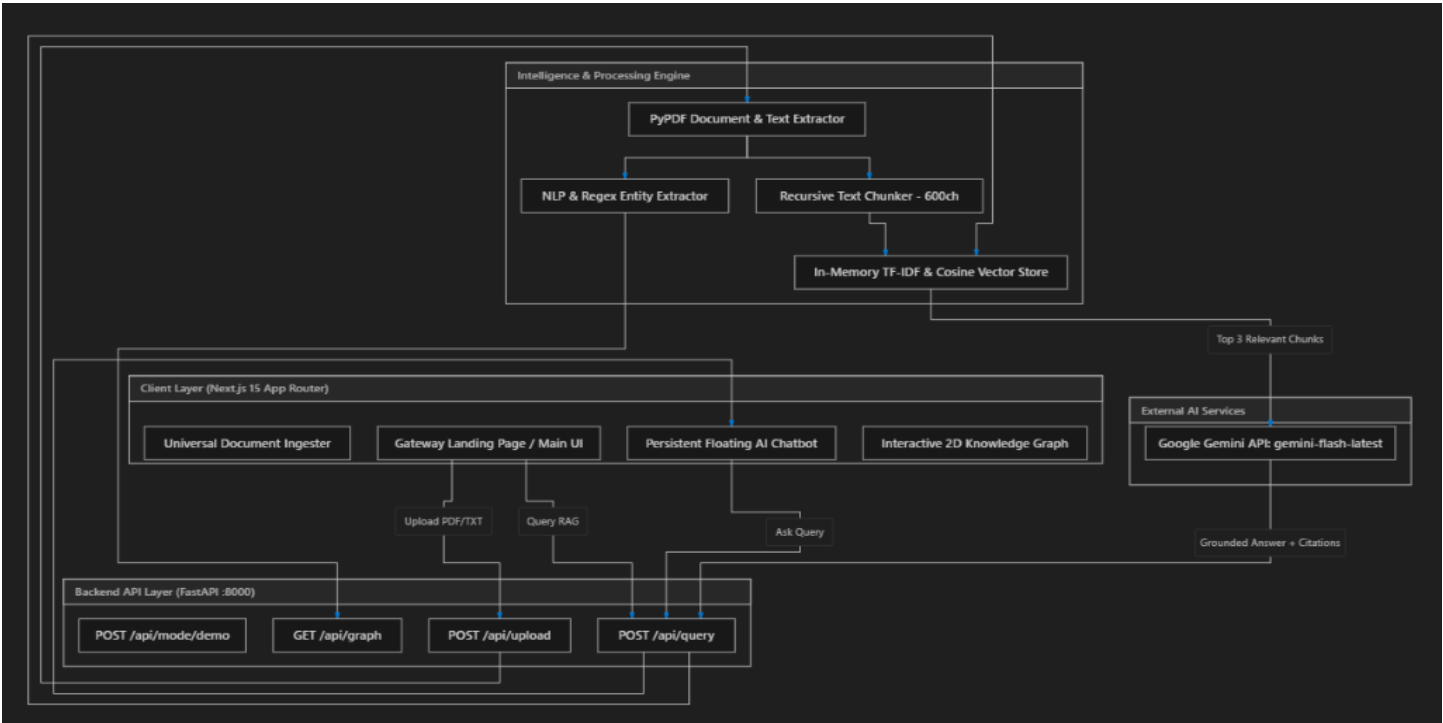


AURA-8 Architecture Diagram & Technical Specification

Economic Times AI Hackathon 2.0 | Problem Statement 8: AI for Industrial Knowledge Intelligence

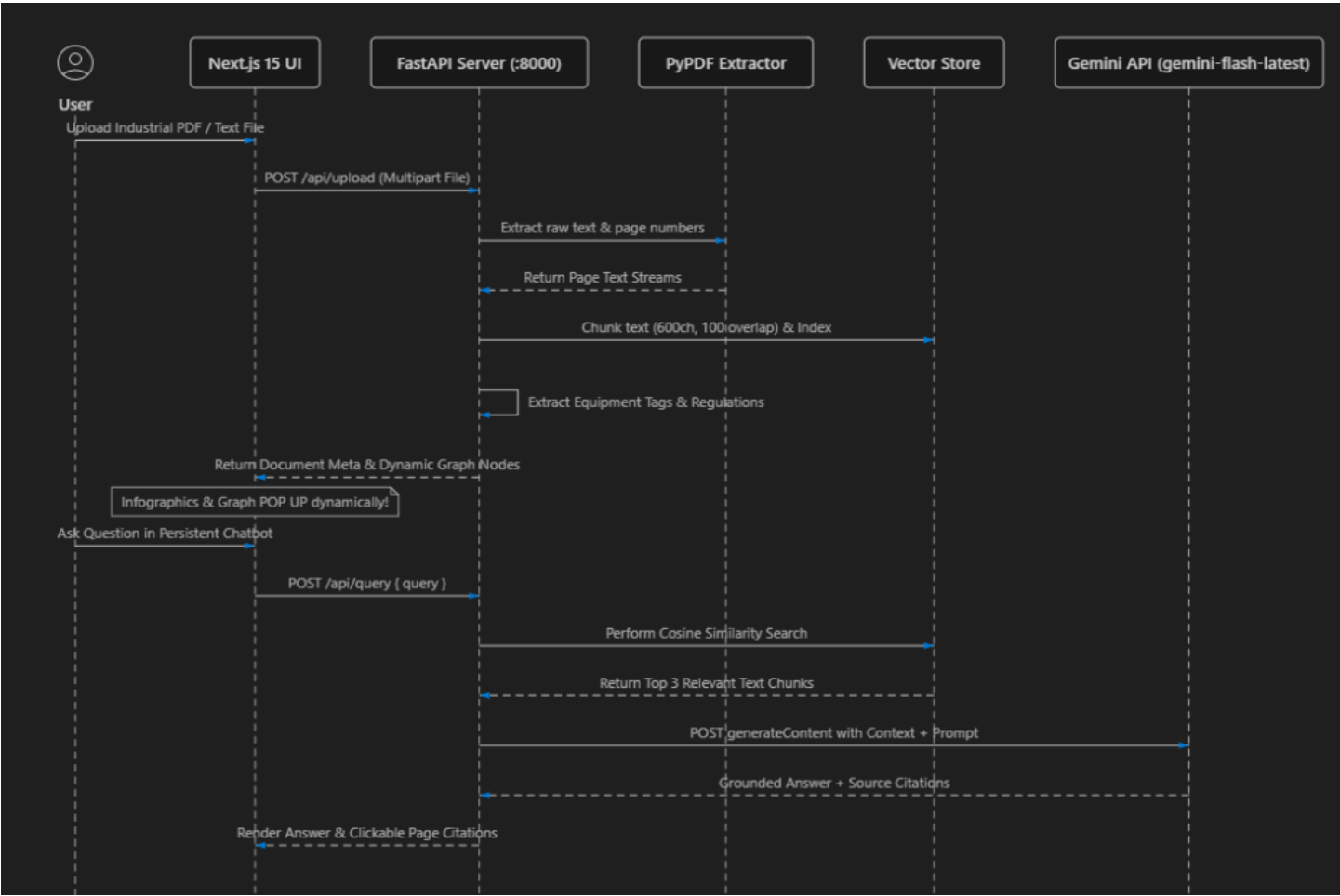
1. High-Level System Architecture Diagram

Complete architectural layer interaction (Client UI → FastAPI/Next.js API → Intelligence Vector Engine → Google Gemini API):



2. Ingestion & RAG Execution Sequence Flowchart

Detailed step-by-step sequence diagram for document upload and grounded RAG query processing:



3. Technology Stack & Component Responsibilities

Subsystem	Technology Stack	Core Responsibility
Frontend UI	Next.js 15, React 19, Tailwind CSS	Gateway landing page, persistent floating chatbot, interactive Canvas 2D graph
RAG Backend	Next.js API Routes & Python FastAPI	Same-origin /api/upload and /api/query endpoints with zero CORS issues.
LLM Engine	Google Gemini (gemini-flash-latest)	Synthesizing grounded responses, page citations, and executive summaries.
PDF Parser	pdf-parse (Node) & PyPDF (Python)	Decompressing PDF streams, extracting equipment tags (P-101A) and OISD co